

Bone regeneration in critical-size calvarial defect using functional biocompatible osteoinductive herbal scaffolds and human umbilical cord Wharton's Jelly-derived mesenchymal stem cells

Shivaji Kashte^a, Rohit Dhumal^b, Pradip Chaudhary^c, R.K. Sharma^d, Vikas Dighe^{e,*,**}, Sachin Kadam^{a,f,*}

^a Department of Stem Cells & Regenerative Medicine, Centre for Interdisciplinary Research, D. Y. Patil Education Society (Institution Deemed to be University), Kolhapur, 416006, India

^b Livestock Development Officer, Department of Animal Husbandry, Government of Maharashtra, Aurangabad, Maharashtra, India

^c Division of Animal Oncology, Advanced Centre for Treatment, Research and Education in Cancer (ACTREC), Navi Mumbai, India

^d D. Y. Patil Medical College, D. Y. Patil Education Society (Deemed to be University), Kolhapur, 416006, India

^e National Centre for Preclinical Reproductive and Genetic Toxicology, National Institute for Research in Reproductive Health (NIRRH), Parel, Mumbai, 400012, India

^f Advancells Group, NOIDA, Uttar Pradesh, 201301, India

ARTICLE INFO

Keywords:

Critical size calvarial bone defect
Scaffold
Graphene oxide
Cissus quadrangularis
Human umbilical cord-derived mesenchymal stem cells
Bone tissue engineering

ABSTRACT

Critical size calvarial bone defects and their repair is the major challenge in orthopaedic surgery. Bone tissue engineering using varied combinations of scaffolds, stem cells, and growth factors is an emerging choice of treatment for variety of bone defects. Present study aimed to compare and evaluate the potential of poly ε-caprolactone (PCL), PCL- graphene oxide (GO), PCL-GO-*Cissus quadrangularis* (CQ), and human umbilical cord-derived mesenchymal stem cells (hUCMSCs) seeded PCL (PCL- hUCMSCs), PCL-GO-CQ-hUCMSCs scaffolds to heal critical size calvarial bone defect in the rat models. Healthy adult Wister female rats (N = 30) with average body weight of 350 ± 30 g were divided to 6 groups with five animals in each group. Single critical size calvarial defects of 8 mm were created in the skull of each rat and same were treated with respective scaffolds. The outcome of the treatments was evaluated at 6 weeks and 12 weeks in terms of weight, haematological parameters and biochemical parameters for biocompatibility. New bone regeneration/healing was analysed using digital radiography and micro-computed tomography. Quality of bone formed was analysed by bone mineral density. Histological analysis of the bone tissues was performed using an optical microscope. All the scaffolds were biocompatible and there was no adverse effect of these scaffolds on animals. Higher bone regeneration was observed after 12 weeks of transplantation than 6 weeks. However, PCL-GO-CQ-hUCMSCs scaffolds exhibited highest bone regeneration 12 weeks post-transplantation. The unique combination of PCL-GO-CQ-hUCMSCs scaffold proved to be beneficial in bone regeneration by aiding nearly complete healing of defect site. Further studies with PCL-GO-CQ-hUCMSCs could pave the way for human clinical trials.

1. Introduction

Critical size calvarial bone defects and their repair is a major treatment challenge in orthopaedic surgery, as this smallest osseous defect in a particular bone shows less than 10 % of spontaneous bone regeneration during lifetime of that animal. These critical size defects that arise due to severe trauma affect quality of life of the patients. It also entails long and postoperative therapy expenses and causes large surgical,

socio-economic problems [1,2].

Although, autografts and allografts are widely used in repair of osseous defects, their use is limited due to tissue availability, disease transmission, donor morbidity and high cost [3]. Bone tissue engineering is an emerging treatment of choice in the repair of bone defects with different combinations such as scaffolds, stem cells, and growth factors. However, scaffolds provide a temporary framework for cell adhesion, growth and their proliferation, and differentiation [4,5]. Various

* Corresponding author at: Advancells, NOIDA, Uttar Pradesh, 201301, India.

** Corresponding author.

E-mail addresses: dighevikas@hotmail.com (V. Dighe), Kadamsachin@gmail.com (S. Kadam).

<https://doi.org/10.1016/j.mtcomm.2021.102049>

Received 21 April 2020; Received in revised form 2 October 2020; Accepted 8 January 2021

Available online 16 January 2021

2352-4928/© 2021 Elsevier Ltd. All rights reserved.

polymers like poly ϵ -caprolactone (PCL), poly-D, L-lactic acid (PDLLA) [6], poly (L/DL-lactide) (PLDL) [7], PLLA [8], poly (DL-lactic-co-glycolic acid) (PLGA) [9] are used for fabrication of Scaffolds [10]. However, FDA approved PCL prepared using electrospinning method was found to be biocompatible, biodegradable and have extensively used in tissue engineering applications. In case of bone tissue engineering, PCL has been studied alone and/or in composite form along with Hydroxyapatite (HA) [11], gelatin hybrid nanofibrous membranes [12], Poly(1, 4-butylene adipate-co-polycaprolactam) (PBAPCL)-HA composite [13]. Surface modification, the other important aspect in tissue engineering, is helpful for cell attachment and proliferation [14]. Therefore, graphene oxide and *Cissus quadrangularis* (CQ) were used in the present study. Graphene and its oxide GO due to their high nano-patterned surface area, mechanical, electroconductive and anti-bacterial properties [15] have emerged as new choice of material for bone tissue engineering.

CQ plant has immense medicinal properties with reported antimicrobial and antioxidant activity [16]. Methanolic extract of CQ revealed a complete and faster bone healing in an experimentally fractured radius-ulna of the dog [17]. The bioactive ingredients of CQ viz. calcium, phosphorous, ketosteroids, phenols, tannins, carotene and vitamin C triterpenes, β -sitosterol, α and β -amyryns, are known to contribute for proliferation and differentiation of mesenchymal stem cells (MSCs) into osteoblasts and bone tissue formation via wnt-LRP5- β -catenin or MAPK dependent pathway [18]. It was also reported that PCL-CQ-HA nanofibrous scaffolds had appropriate surface roughness for the osteoblast adhesion, proliferation, and mineralization, making them potential biocompatible material for bone tissue engineering [19].

Unlike other tissues, bone has unique properties of regenerating subcritical size defects by recruitment of MSCs from the bone marrow and surrounding vasculature. External chemical stimuli by ascorbic acid, β -glycerophosphate, and dexamethasone helps MSCs to differentiation into osteoblasts and osteoclasts [20]. Along with bone marrow and peripheral blood other adult tissues like dental pulp, umbilical cord, adipose tissue are becoming popular MSCs sources. Human Umbilical cord matrix or Wharton jelly derived MSCs are an increasingly popular foetal origin cells since they are abundantly available medical waste with no ethical issues associate [21]. Further, they demonstrated superior capacity for osteogenic differentiation as compared to adult sources [22].

The present study was undertaken to evaluate the *in vivo* potential of the previously prepared plain PCL, and layer-by-layer surface modified PCL scaffold using only GO and GO in combination with CQ with or without human umbilical cord derived MSCs (hUCMSCs) (PCL-hUCMSCs, PCL-GO, PCL-GO-CQ, PCL-GO-CQ-hUCMSCs) to heal critical-size calvarial bone defect of 8 mm in the rat models. Along with bone tissue regeneration *in vivo* biocompatibility, safety host tissue response of local and system tissue to transplanted scaffolds have also been evaluated.

2. Materials and methods

2.1. Preparation of scaffolds by layer by layer method

Scaffolds for the study were prepared by electrospinning method and their surface was modified using layer-by-layer method as described previously by Kashte et al. [23]. Briefly, nanofibrous scaffold of PCL were prepared by electrospinning 10 % w/v PCL solution. The solution of GO (1 mg/mL) and a pure callus cultured extract of CQ (1 mg/mL) were prepared by dispersing components in distilled water through sonication. These solutions were used to modify the surface of PCL scaffolds fabricated by electrospinning. Preparation of PCL-GO, PCL-GO-CQ scaffolds were carried out by 2 min repetitive dipping of the PCL scaffolds in GO, and CQ solution respectively and air dried. Total 60 repetitive dipping and drying cycles were carried out for optimum deposition. In our earlier study, these prepared scaffolds were characterised for physical, mechanical and biological properties including *in vitro* biocompatibility, and bone differentiation capability using

hUCMSCs and were found to be potential for bone tissue engineering applications [23].

2.2. Animals and housing

The animal experiments were executed with prior approval from the Institutional Animal Ethics Committee of the National Institute for Research in Reproductive Health (NIRRH) (approval no. NIRRH/IAEC/11/18).

The animal study was carried out in accordance with the set guidelines of the Committee for the Purpose of Control and Supervision of Experimental Animals (CPCSEA), India. A total of 60 Wister female rats of 6 to 8-week-old with average body weight (BW) of 350 ± 30 g were used in this study. All animals were housed at experimental animal house facility at NIRRH. They were accommodated in polypropylene cages at monitored temperature of 23 ± 1 °C, humidity of 55 ± 5 % and 14 h light and 10 h dark cycle. Sterile corn cobb was used as bedding material that was replaced every 3 days. During the study, soy-free, pelleted feed and filtered drinking water *ad libitum* were given to the animals.

2.3. Study design and surgical procedures

The study consists of 6 groups ($n = 10$) based on type of transplanted scaffold namely PCL, PCL-hUCMSCs, PCL-GO, PCL-GO-CQ and PCL-GO-CQ-hUCMSCs, and Sham. Each group was further divided into two subgroups based on observation period i.e. 6 week and 12 weeks. Prior to implantation, all the scaffolds were sterilized by ethylene oxide (EtO). The hUCMSCs were isolated from human umbilical cord Wharton's Jelly by digesting with a mixture of enzymes viz. Collagenase Type IV and Dispase (7:1 v/v) and Trypsin-EDTA. The isolated hUCMSCs were cultured in the Dulbecco's Modified Eagle's Medium: Ham's F12 (DMEM: HF12, 1:1) medium supplemented with 10 % Foetal calf serum and penicillin (100 units/mL) and streptomycin (100 μ g/mL). The cells were incubated for 48 h at 37 °C, 5% CO₂ [23] until 80 % confluency. These isolated hUCMSCs were seeded onto respective scaffolds at cell density 4×10^5 cells/per scaffold and further cultured for next 24 h. The scaffolds with cells were then washed with sterile PBS to remove traces of the culture medium.

Rats from all the groups were anesthetised using ketamine HCL (60 mg/kg, Intas pharma Ltd., India) and xylazine (7 mg/kg, Themis Medicare Ltd., India). The site of the surgery was clean-shaven and sterilized with betadine (G.S. Pharmbutor Pvt. Ltd., India) and 70 % ethanol. The soft tissue and periosteum were uplifted to reveal the calvarial surface on the sagittal suture line. On the sagittal suture line of parietal bones, the critical size bone defects (8 mm diameter) were made using a trephine bur along with a dental hand piece and micromotor, with utmost precaution to avert damage to both the sagittal suture and the brain. To avoid heat generation, sterile saline was constantly irrigated during defect crafting. The scaffolds were implanted into the defect site in respective animal groups for 6 weeks and 12 weeks. The Sham group was kept empty without any scaffold. The pericranium and skin were sealed using topical skin adhesive (Dermabond Advanced®, J&J, India).

After surgery, each rat received an *i.p.* injection of gentamycin (3 mg/kg body weight), ampicillin and Cloxacillin (20 mg/kg body weight) and diclofenac sodium (0.5 mg/kg body weight) in addition to topical ointment (Soframycin®, Aventis Pharma Ltd, Pune, India) and was caged individually. These animals were examined for clinical indications and symptoms. The weights of animals before and after surgery (transplantation of scaffold) were taken at 6 and 12 weeks' time points. Also, the weight of essential organs was taken after the surgery at both 6 weeks and 12 weeks.

2.4. Blood collection, haematology and serum biochemistry

The clinical pathology and examination were performed for any inflammatory reactions after implantation of scaffolds, through a complete blood count [CBC, Abacus (Diatron MI PLC, Hungary) haematology analyser]. After 6 weeks and 12 weeks of transplantation, animals were bled from retro-orbital sinus and 1 mL blood was collected in two vials, with and without heparin. The heparin (2IU/mL) containing vial was utilized for evaluation of haematological parameters like haemoglobin (Hb) (g/dL), packed cell volume (PCV; %), total red cell count (RBC), total white cell count (WBC; (lymphocytes, monocytes, neutrophils, eosinophiles, and basophiles) were calculated morphologically by blood smears), absolute erythrocyte indices, differential WBC count. The other vial without heparin was utilized for evaluation of serum biochemistry parameters such as serum glutamic pyruvic transaminase (SGPT), serum glutamic oxaloacetic transaminase (SGOT), alkaline phosphatase (ALP), creatinine and urea were evaluated on ERba Chem 7 (Erba Mannheim, Germany) semi-autoanalyzer using commercial reagent kits. The concentrations of Osteocalcin (OCN) were measured using an enzyme-linked immunoassay kit (Invitrogen, USA). The coating buffer was used to coat the microwell plate prior to addition of samples. The blocking buffer was used to block the other active sites. The samples then incubated with primary antibody Antihuman Osteocalcin for one hour at room temperature and then with HRP conjugated secondary antibody for 1 h at room temperature. The substrate 3,3',5,5'-tetramethylbenzidine (TMB) was added into each well and incubated for 30 min. The reaction was then ended by adding stop solution. The absorbance was measured at 450 nm. The lower detection limit of the OCN concentrations was 1.0 ng/mL.

A differential WBC count (lymphocytes, monocytes, neutrophils, eosinophiles, and basophiles) was calculated morphologically by blood smears. CBC was performed using Abacus (Diatron MI PLC, Hungary) haematology analyser, while serum biochemistry evaluation was done on ERba Chem 7 (Erba Mannheim, Germany) semi-autoanalyzer using commercial reagent kits.

2.5. X-ray and Micro-CT analysis

Rat calvaria was harvested from rat skull and imaged using a digital X-ray machine at 30 kV and 3 mA for 1.5 min. Rat skulls after euthanasia were fixed in ethanol and washed before scanning on an animal micro-computed tomography (CT) scanner (FLEX Triumph micro-PET/SPECT/CT scanner, Gamma Medica-Ideas, USA) using an X-ray tube potential of 75 kV and a tube current of 100 μ A. The scanner was equipped with an X-ray source and a detector system. It turned around the bed having the calvaria. The data was acquired over 360° producing 1024 views. Each frame was exposed for 2000 milli sec. The total image acquisition was 45 min and images were reconstructed using a modified cone-beam algorithm with an isotropic voxel spacing of $0.045 \times 0.045 \times 0.045 \text{ mm}^3$ over a region of interest.

Furthermore, micro-CT scan was also performed on rat skulls isolated from the Sham control as the negative control. Scan parameters used were 80 kVp, 0.45 mA and 180 mAs. All the factors and circumstances were kept constant for all samples.

Newly regenerated bone quality analysis in terms of bone mineral density (BMD) (mg/mm^2) was performed using MicroView software (Parallax Innovations, Inc., Canada) to evaluate the quality of bone formed after 12 weeks of transplantation of scaffolds. The harvested calvaria were scanned for BMD in the region of interest (ROI). The normal calvaria were scanned to have BMD control values for comparison with test groups.

2.6. Histological analysis

Post-harvest, cranium samples were fixed in neutral buffered formalin (NBF) at 4 °C for 48 h. The calvaria were decalcified in HCl for

3 weeks were processed and embedded in paraffin. The blocks were trimmed and sectioned with a microtome (Leica RM2156; Leica, Germany) equipped with a tungsten carbide knife. The sections were stained with haematoxylin and eosin (H&E) and the histological analysis of the bone tissues was performed using an optical microscope (Carl Zeiss, Germany).

3. Statistical analysis

Statistical analysis of data was evaluated by Origin Pro 8.5 software (Origin Lab Corporation, Massachusetts, U.S.A.). Data were presented as the Mean $\pm$ standard deviation. Statistical analysis was performed by using a one-way analysis of variance (ANOVA). A p value of < 0.05 was considered to be statistically significant.

4. Results and discussion

A nanofibrous PCL scaffolds were prepared by electrospinning. These scaffolds were surface modified by layer-by-layer method using GO and CQ. The porous and hydrophilic nature of these scaffolds along with mechanical stability helped *in vitro* cellular proliferation. In detailed preparation, characterisation and *in vitro* biocompatibility of the prepared scaffolds have been reported previously [23]. These results indicate that the surface modification of PCL scaffolds with GO and CQ have improved osteoinductive [23,24] and osteoconductive [24,25] properties as well as they helped in spontaneous differentiation of hUCMSCs to osteoblast without osteogenic differentiation media. The present study particularly deals with the *in vivo* biocompatibility of these scaffolds along with quick healing of critical size calvarial bone defect of 8 mm in the rat models.

Gross observation of animals' post-surgery in the entire control group did not show any infection. Implanted scaffold did not produce any surgery or implant related abnormalities throughout the study period of 12 weeks.

4.1. Animal weight and organ weight

The weights of animals before and after 6- or 12-weeks surgery (transplantation of scaffold) were in normal range with no significant weight loss (Supplementary Information). Also, the weight of essential organs i.e. Liver, Kidney, Heart, Spleen Lungs and their respective relative weights to the body weights (Supplementary Information) were in the normal range. Similar body weights, organ weights and their relative weights were mentioned elsewhere [26–28]. The body weight, weight of essential organs and their respective relative weight in the normal range showed that there was no adverse effect of the transplanted scaffolds on these organs or on the animal itself.

4.2. Blood collection, haematology and serum biochemistry

The blood haematological parameters were evaluated to assess any possible infections due to implantation of scaffolds in rat skull defects. CBCs and differential leukocytes counts among all the groups were within the normal range after 6 and 12 weeks (Supplementary Information). (Normal range for Hb: 11–19.2 g/dL, MCV: 46–70 fL, PCV: 35–60 %, MCH: 14–18 Pg, MCHC: 30–35 g m/dL, RBC: $7\text{--}12.5 \times 10^6/\text{mm}^3$, WBC: $0.96\text{--}9 \times 10^3/\text{mm}^3$, neutrophils: 8–55 %, monocytes: 0–12 %, eosinophiles: 0–4.7% lymphocytes: 60–85 %, platelets: $360\text{--}720 \times 10^3/\text{mm}^3$) [29]. This indicated that scaffold transplantation did not evoke an inflammatory response.

Similar results were observed when poly (L-lactic acid)-hydroxyapatite- gelatin and poly (L-lactic acid)-hydroxyapatite scaffolds were implanted in critical-size calvarial bone defects for 6 weeks and 10 weeks [30]. Also, there was no inflammatory reaction when PCL-hydroxyapatite and PCL-hardystonite scaffolds were implanted for 6 weeks and 12 weeks in rat calvarial defects [31]. Also, a similar range of

haematological parameters were mentioned elsewhere [27].

The biochemical parameters (Fig. 1) were examined to check the efficacy of essential organs of the animal body such as liver and kidney. In serum biochemistry liver enzymes SGPT (Normal range: 25–120 IU/L) and SGOT (Normal range: 45–110 IU/L) were found to be in the normal range. SGPT and SGOT are markers of liver functioning and their normal range signifies the proper working of the liver. Urea and Creatinine are markers of renal functioning. Serum urea (Normal range: 12–40 mg/dL) and Creatinine levels (Normal range: 0.2–1 mg/dL) among all the groups were within the normal range. This signifies healthy state of the renal system. Similar results were observed when poly (L-lactic acid)-hydroxyapatite-gelatin and poly (L-lactic acid)-hydroxyapatite scaffolds were implanted in critical-size calvarial bone defects for 6 weeks and 10 weeks [30]. Also, a similar range of urea and creatinine levels were mentioned elsewhere [27,28]. No adverse inflammatory reactions and any infection was observed in any animal post transplantation, this proves the biocompatible nature and bio-acceptability of scaffolds.

Osteocalcin (OCN) (Fig. 2) levels were higher at 6 weeks and 12 weeks in all the groups than control normal rat (Normal range: 2–9 ng/mL). OCN concentrations are indicators of active osteoblasts that help to form new bone. The bone-forming cells osteoblasts were active at 6 weeks and 12 weeks in all the groups. OCN concentrations were slightly lower in 12 weeks than in 6 weeks for PCL-GO-CQ and PCL-GO-CQ-hUCMSCs groups. The rise of OCN concentrations up to 10 weeks and then sharp decline in concentration was observed. Similar fluctuations and lower concentrations of OCN were found in bone fracture healing patients after 10 weeks of healing [32,33] (Fig. 3).

The ALP concentration (Normal range: 220–350 IU/L) was lower at 12-week time point among all the scaffold groups as compared to 6 weeks except PCL-GO group (Fig. 2). At the 6 weeks, the PCL-GO-CQ-hUCMSCs group showed the highest concentrations of ALP compared to other groups. ALP is the marker of wound healing, tissue repair and angiogenesis. Both values of OCN and ALP are in sync with each other and it indicated higher tissue repair activities [32,33].

4.3. X-ray and Micro-CT analysis

The digital X-ray images of rat calvaria of implanted groups after 6 weeks and 12 weeks were shown in (Supplementary Figure 15). No bone regeneration was observed in the defects of the Sham group. This could be due to the critical size of the bone defect of 8 mm, and no support for bone healing. Also, there was minimal bone healing in defect implanted with PCL, while in PCL-hUCMSCs and PCL-GO there was slight bone healing, and there was highest bone healing in PCL-GO-CQ groups after 6 weeks and 12 weeks of transplantation. However, after 6 weeks and 12 weeks transplantation, the PCL-GO-CQ-hUCMSCs group was showing the maximum amount of radiolucent bone formation in the calvarial defect site.

The 3D reconstructed images of bone calvaria from all the groups after 6 weeks and 12 weeks of implantation are shown in Fig. 4. It was observed that Sham control group has the least bone healing after 6 weeks and 12 weeks of the study period as the defect was left untreated. The other groups PCL-hUCMSCs, PCL-GO were showing slightly higher bone growth after 12 weeks than 6 weeks of transplantation. The PCL-GO-CQ group shows higher bone growth as compared to these groups. The highest amount of bone growth was observed in the PCL-GO-CQ-hUCMSCs group after 6 weeks and 12 weeks of transplantation. There was an almost complete closure of calvarial bone defect after 12 weeks. This shows that hUCMSCs seeded into the PCL-GO-CQ scaffolds adhere well proliferated and differentiated into osteoblasts that could boost new bone formation. While in PCL-hUCMSCs, the cells could have adhered but not differentiated into osteoblasts, in PCL-GO-CQ-hUCMSCs group, the synergistic effect of GO and CQ created a niche for MSCs.

Bone mineral density (BMD) is the indicator of new bone formation. It was calculated at the defect site after implantation and was analysed by MicroView. Fig. 5 showed the BMD values after implantation of 12 weeks. The lowest BMD values were of Sham control (29.5 mg/cc) indicated minimum bone formation. The highest BMD values were of PCL-GO-CQ-hUCMSCs group (310 mg/cc) as compared to PCL (43.5 mg/cc), PCL-hUCMSCs (84.5 mg/cc), PCL-GO (84 mg/cc), PCL-GO-CQ (100 mg/cc) groups after 12 weeks of implantation. Normal bone

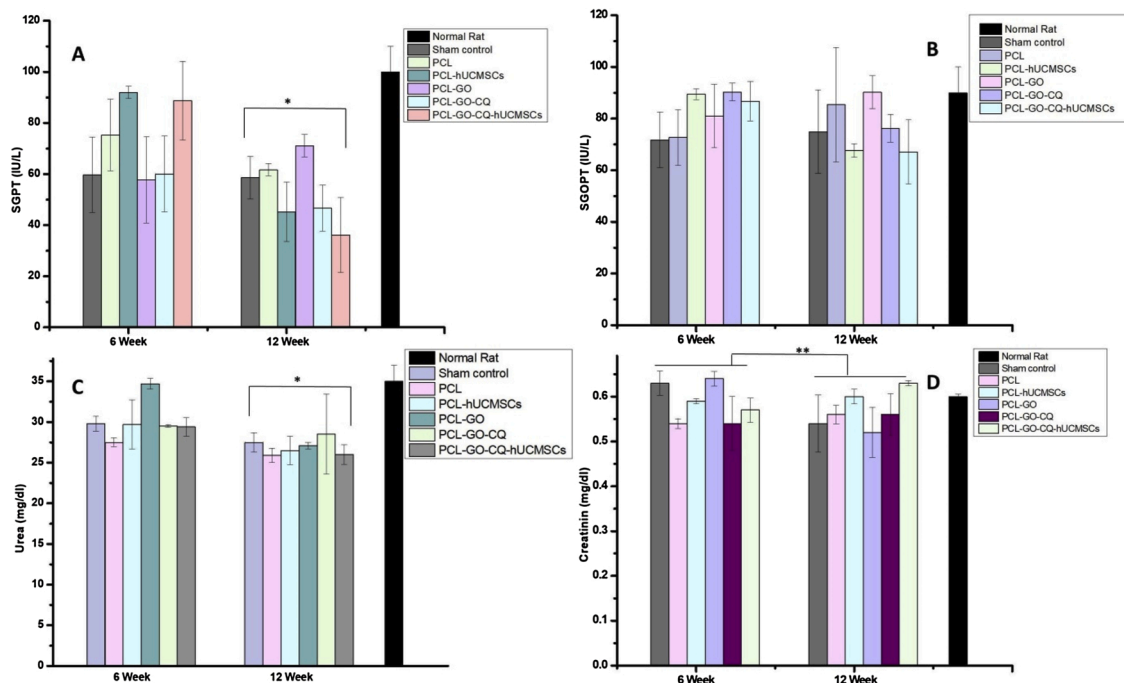


Fig. 1. Biochemical Parameters: A) SGPT, B) SGOT, C) Urea and D) Creatinine values described for 6 weeks and 12 weeks. SGPT values were in the normal range. There was a significant difference in 12 weeks among the groups ($p < 0.05$). SGOT values were in the normal range. There was no significant difference among the groups ($p > 0.05$). Urea values were in the normal range. There was a significant difference in 12 weeks among the groups ($p < 0.05$). Creatinine values were in the normal range. There was a significant difference between the 6 weeks and 12 weeks group ($p < 0.05$).

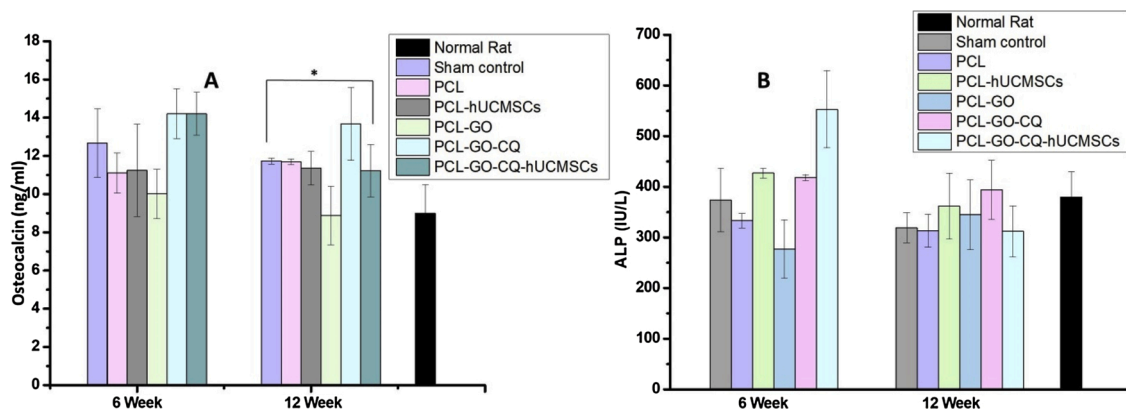


Fig. 2. A) Osteocalcin (OCN) and B) ALP values described for 6 weeks and 12 weeks. OCN values were in the normal range. There was a significant difference among the 12-week groups ($p > 0.05$). ALP values were in the normal range. There was no significant difference among the groups ($p > 0.05$).

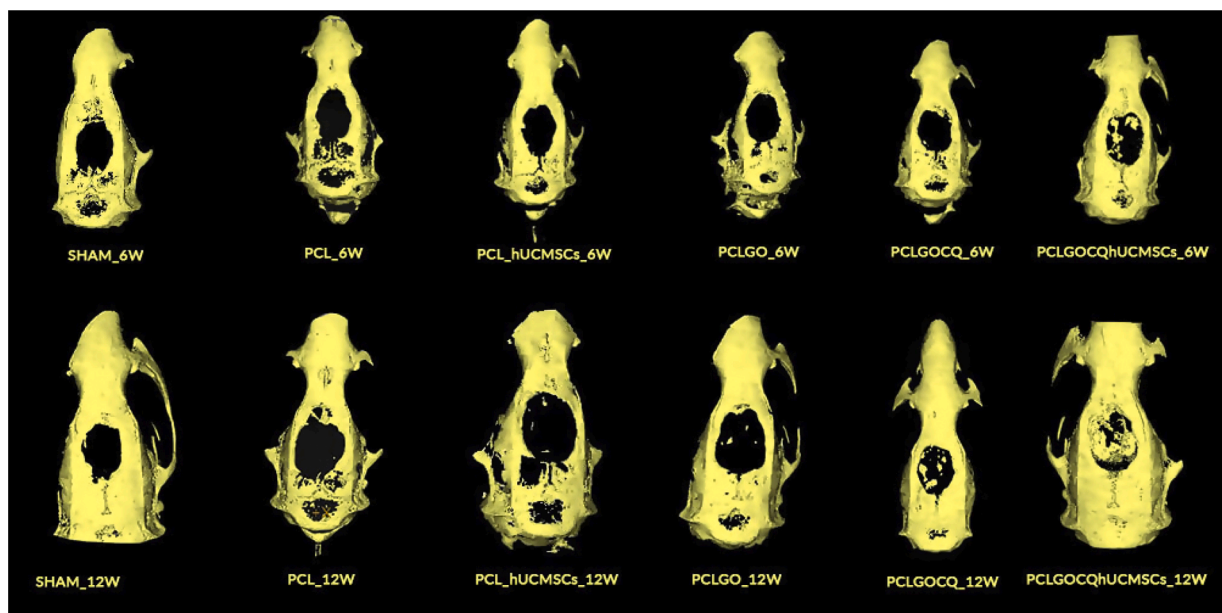


Fig. 3. Reconstructed 3D micro-CT images of rat calvaria implanted with scaffolds after 12 weeks. The highest bone regeneration was found in the PCL GO CQ hUCMSCs scaffolds after 6 weeks and 12 weeks of transplantation.



Fig. 4. Bone mineral densities of Scaffolds after 12 weeks of implantation into rat calvaria. No significant difference ($p > 0.05$) observed among groups at 12 weeks of implantation.

BMD (625 mg/cc) was highest as compared to all the implanted groups. These results were comparable to digital X-ray and micro-CT images. When poly (L-lactic acid)-hydroxyapatite-gelatin and poly (L-lactic

acid)-hydroxyapatite scaffolds were implanted in critical-size calvarial bone defects of rats for 10 weeks, it showed BMD values of ~ 110 mg/cc and ~ 110 mg/cc respectively while for sham control and normal bone was ~ 90 mg/cc and ~ 120 mg/cc, respectively [30]. When PCL- hydroxyapatite and PCL-hardystonite scaffolds were implanted for 6 weeks and 12 weeks in rat calvarial defects, BMD values were observed as ~ 60 mg/cc, ~ 80 mg/cc while for sham control and normal bone were ~ 40 mg/cc and ~ 300 mg/cc [31]. After 12 weeks of implantation in critical size bone defect of New Zealand rabbits, control, Chitosan-Poly (vinyl alcohol)- (CPH) and hydroxyapatite coated CPH (CPHH) showed BMD values of ~ 40 mg/cc, ~ 100 mg/cc and ~ 170 mg/cc respectively [34].

4.4. Histological analysis

The histology results (Figs. 5 & 6) were in resemblance with micro CT and radiology results. There was minimal new bone tissue formation, low density of osteoblasts in Sham and PCL group after 6 weeks of transplantation. There was medium new bone tissue formation, and medium density of osteoblasts in PCL-hUCMSCs and PCL-GO group after 6 weeks of transplantation. There was high new bone tissue formation and high density of osteoblasts in PCL-GO-CQ group after 6 weeks of

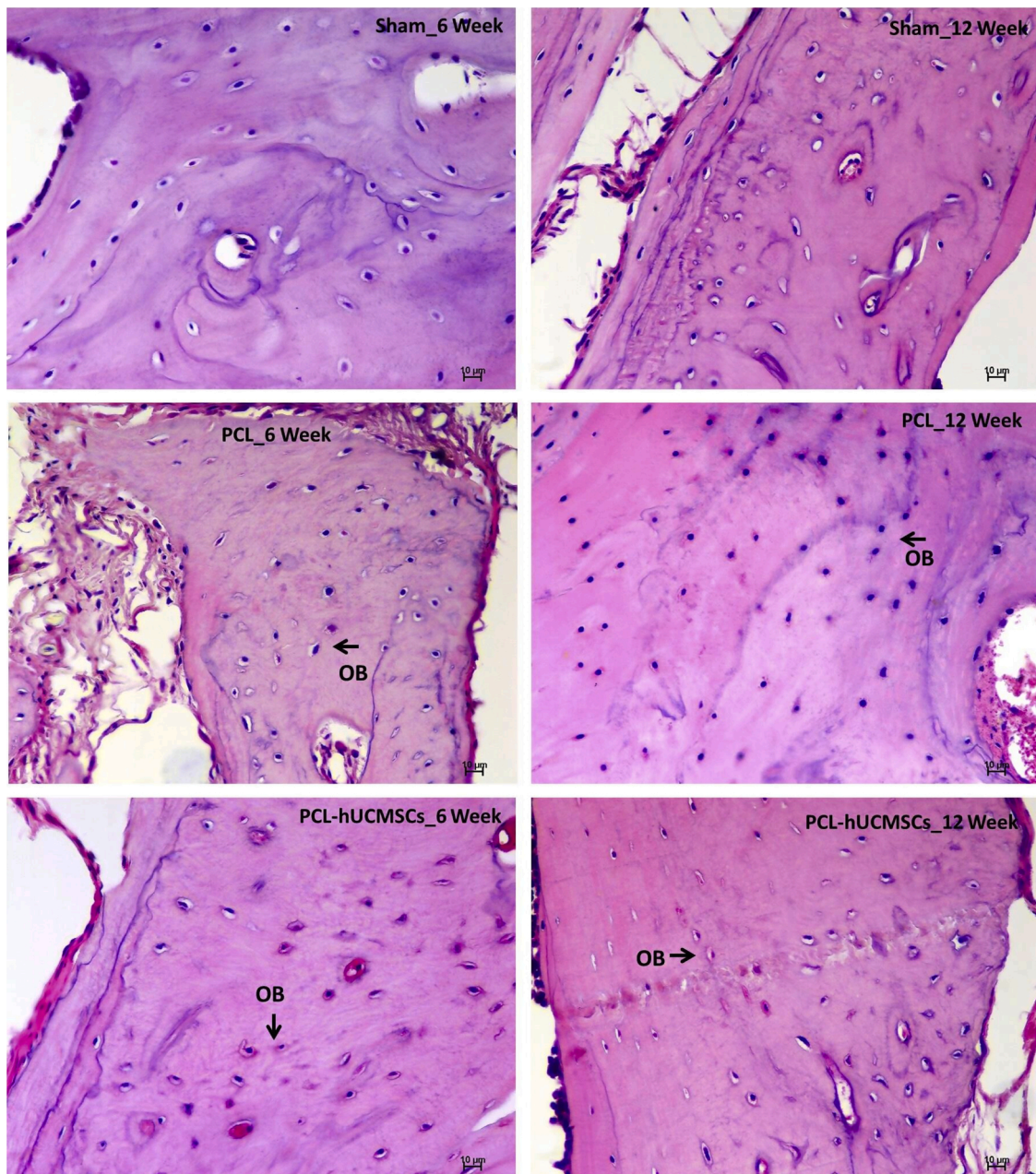


Fig. 5. Histological analyses of calvaria after 6 weeks and 12 weeks of transplantation in Sham, PCL and PCL-hUCMSCs groups. (OB: Osteoblasts; OC: Osteocytes, Scale bar 10 μ m).

transplantation. There was higher new bone tissue formation, a higher density of osteoblasts in PCL-GO-CQ-hUCMSCs group after 6 weeks of transplantation.

There was minimal new bone tissue formation, low density of osteoblasts in Sham and PCL group after 12 weeks of transplantation. There was medium new bone tissue formation, medium density of osteoblasts in PCL-hUCMSCs and PCL-GO group after 12 weeks of transplantation. There was high new bone tissue formation, a higher density of osteoblasts and osteocytes in PCL-GO-CQ group after 12 weeks of transplantation. There was highest new bone tissue formation, a highest density of osteoblasts and osteocytes in PCL-GO-CQ-hUCMSCs group after 12 weeks of transplantation. Also, there was matrix mineralization. It indicates bone remodelling activities.

There was no abnormal development in bone tissues, inflammatory changes or necrosis after 6 weeks of transplantation or 12 weeks of transplantation. Inflammatory cells are the primary source of the signals

during the initial phase of bone healing. However, derivatives of MSC lineage including osteoblasts and osteoclasts are also responsible for the secretion of inflammatory cytokines within 3–7 days of injury and subsequent phases of healing [35,36]. Also, several *in vivo* studies have revealed that neutrophils, macrophages and lymphocytes were not always necessary for tissue repair. Their absence may even accelerate healing [37,38].

Similar kinds of results were mentioned by other researcher. There was increased new bone formation and density of bone-forming cells in rat calvarial defects when BMSCs were seeded on chitosan-alginate-hydroxyapatite scaffolds loaded with BMP-2 as compared to Chitosan-alginate-hydroxyapatite, 12 week of transplantation [39]. When polymethylmethacrylate (PMMA) and platelet gel (PG) were transplanted in radial bone defect of rats for 8 weeks there was an increased activity of osteoblasts, osteoclasts and new bone formation [40]. Histological analysis showed that there were osteoblastic proliferation, osteoid

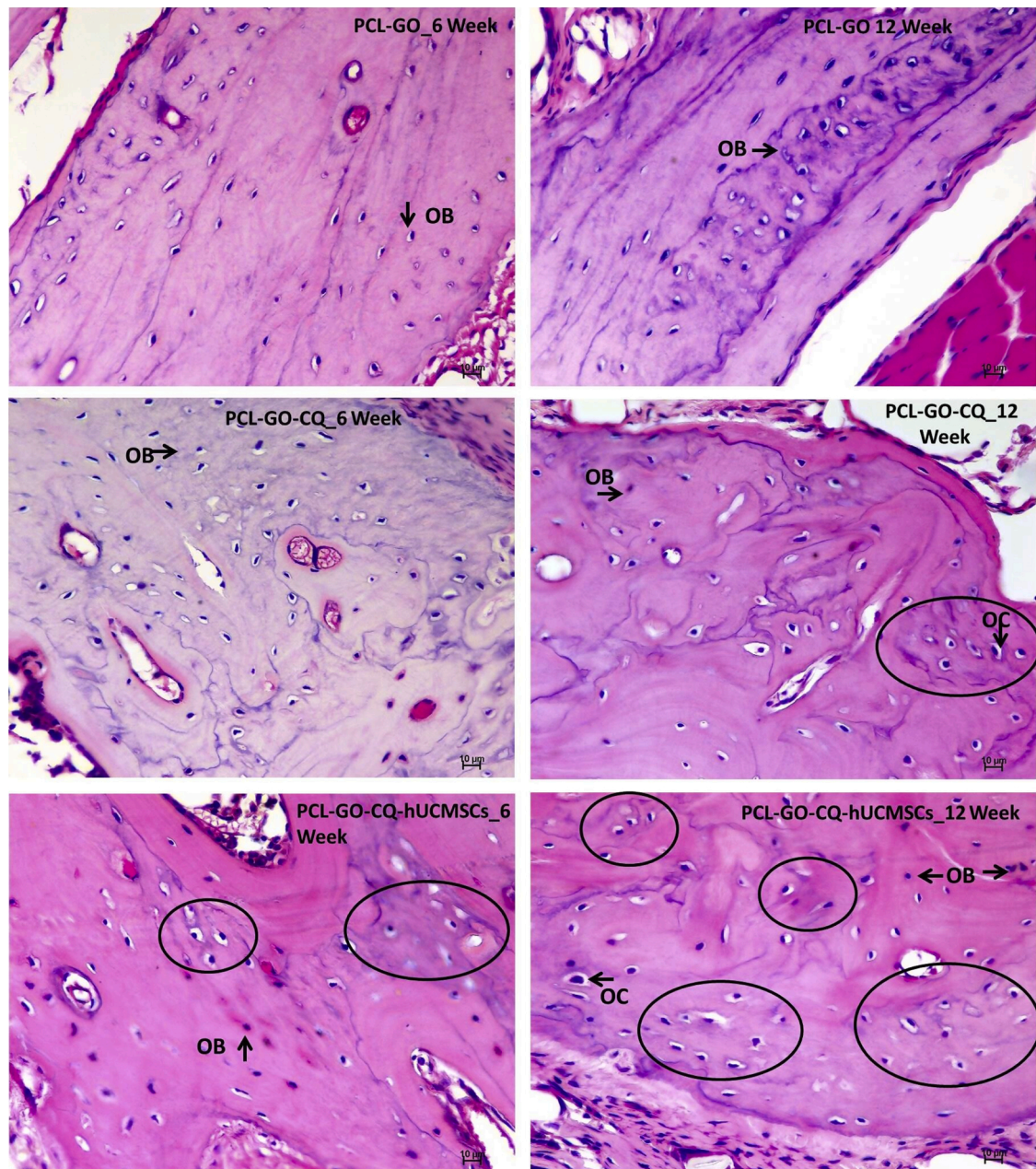


Fig. 6. Histological analyses of calvaria after 6 weeks and 12 weeks of transplantation in PCL-GO, PCL-GO-CQ and PCL-GO-CQ-hUCMSCs. (OB: Osteoblasts; OC: Osteocytes; elliptical areas: matrix mineralization), Scale bar 10 μ m.

matrix, with the presence of osteoclasts when poly (L-lactic acid)-hydroxyapatite-gelatin and poly (L-lactic acid)-hydroxyapatite scaffolds were implanted in critical-size calvarial bone defects for 6 weeks and 10 weeks [30]. Also when PCL- hydroxyapatite and PCL-hardystonite scaffolds were implanted for 6 weeks and 12 weeks in rat calvarial defects, histology revealed there was higher cell activity with the presence of inflammatory cells [31].

The overall new bone tissue formation was higher in 12 weeks of transplantation compared to 6 weeks of transplantation. There was increased new bone tissue formation in hUCMSCs scaffolds as compared to scaffolds devoid of hUCMSCs. The bone formation process in the flat bone-like skull occurs *via* intramembranous direct ossification. There was a slow process of bone formation due to the limited supply of MSCs in flat bones [41]. This could be the main reason, as there is minimal new bone formation in the defect area in all the scaffolds groups except PCL-GO-CQ-hUCMSCs group. In PCL-GO-CQ-hUCMSCs group, the

roughness, porosity of PCL-GO-CQ scaffold allowed hUCMSCs to adhere, grow and proliferate. However, osteoconductive and osteoinductive properties of GO and CQ eventually differentiated hUCMSCs to osteoblasts and filled the bone defect with new bone tissue. The bioactive components present in CQ extract *viz.* calcium, phosphorous, phytoestrogenic steroids [42,43], ketosteroids, phenols, tannins, carotene and vitamin C triterpenes, β -sitosterol, α and β - amyrins, are known to influence early regeneration and mineralization in the process of bone healing. These phytochemicals contribute in proliferation and differentiation of mesenchymal stem cells (MSCs) into osteoblasts and bone tissue formation *via* wnt-LRp5- β -catenin or MAPK dependent pathway [18]. In PCL- hUCMSCs group, however, the PCL alone could not able to proliferate and differentiate hUCMSCs into osteoblasts to regenerate the bone tissues.

5. Conclusions

In the present study, we have investigated the potential of PCL, PCL-GO and PCL-GO-CQ scaffolds for bone tissue engineering. All the scaffolds were biocompatible and not produced any substantial toxic effects on implantation. PCL-GO-CQ scaffolds were superior compared to other scaffolds and regenerated higher bone tissue. But seeding of hUCMSCs into the PCL-GO-CQ scaffolds showed highest amount of bone regeneration at 12 weeks of time point compared to PCL, PCL- hUCMSCs, PCL-GO, PCL-GO-CQ. The substantial regeneration of bone tissue at the defect site post 12 weeks of surgery using hUCMSC-PCL-GO-CQ scaffold could be due to the synergistic effect of GO and CQ. Our earlier *in vitro* studies have demonstrated spontaneous differentiation of hUCMSCs to osteogenic lineage when seeded on to the PCL-GO-CQ scaffold. This was further confirmed in the present *in vivo* study where, hUCMSCs along with PCL-GO-CQ scaffold was able to regenerate new bone effectively in the defect site. Further studies in large animals with this unique combination of PCL-GO-CQ-hUCMSCs scaffolds could pave the way for human clinical trials and could be a potential bone healing material.

Authors contribution

S. Kashte has performed the experiments, interpreted the data and wrote the manuscript. Rohit Dhumal has performed the animal surgery. Pradip Chaudhary has performed mCT analysis. R.K. Sharma has conceived the experiments. Vikas Dighe has conceived the experiments, S. Kadam has conceived the experiments, contributed to the interpretation of the data and writing the manuscript.

Funding

The animal study was funded by intramural grant from Advancells Group.

Declaration of Competing Interest

The authors declare no competing interests.

Acknowledgements

Author would like to acknowledge the University Grant Commission (UGC), Government of India, New Delhi for doctoral fellowship to Dr Shivaji Kashte.

Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.mtcomm.2021.102049>.

References

- W. Wang, K.W.K. Yeung, Bone grafts and biomaterials substitutes for bone defect repair: a review, *Bioact. Mater.* 2 (2017) 224–247.
- S. Stewart, S.J. Bryant, J. Ahn, K.D. Hankenson, Bone Regeneration, *Transl. Regen. Med.*, Elsevier Inc., 2015, pp. 313–333.
- T. Kaur, A. Thirugnanam, K. Pramanik, Effect of carboxylated graphene nanoplatelets on mechanical and in-vitro biological properties of polyvinyl alcohol nanocomposite scaffolds for bone tissue engineering, *Mater. Today Commun.* 12 (2017) 34–42.
- T.W.B.G.F. Muschler, Bone graft materials: an overview of the basic science, *Clin. Orthop. Relat. Res.* 31 (2000) 10–27.
- N.A. Pattanashetti, S. Biscaia, C. Moura, R. Mitchell, Development of novel 3D scaffolds using BioExtruder by the incorporation of silica into polycaprolactone matrix for bone tissue engineering, *Mater. Today Commun.* 21 (2019) 1–12.
- A. Alves, A.R.C. Duarte, J.F. Mano, R.A. Sousa, R.L. Reis, PDLLA enriched with ulvan particles as a novel 3D porous scaffold targeted for bone engineering, *J. Supercrit. Fluids* 65 (2012) 32–38.
- I. Rajzer, E. Menaszek, R. Kwiatkowski, W. Chrzanowski, Bioactive nanocomposite PLDL/nano-hydroxyapatite electrospun membranes for bone tissue engineering, *J. Mater. Sci. Mater. Med.* 25 (2014) 1239–1247.
- X. Liu, P. Ma, Polymeric scaffolds for bone tissue engineering, *Ann. Biomed. Eng.* 32 (2004) 477–486.
- J. Qian, W. Xu, X. Yong, X. Jin, W. Zhang, Fabrication and in vitro biocompatibility of biomimetic PLGA/nHA composite scaffolds for bone tissue engineering, *Mater. Sci. Eng. C Mater. Biol. Appl.* 36 (2014) 95–101.
- S. Kashte, A.K. Jaiswal, S. Kadam, Artificial bone via bone tissue engineering: current scenario and challenges, *Tissue Eng. Regen. Med.* 14 (2017) 1–14.
- B. Chuenjitkuntaworn, T. Osathanon, N. Nowwarote, P. Supaphol, P. Pavasant, The efficacy of polycaprolactone / hydroxyapatite scaffold in combination with mesenchymal stem cells for bone tissue engineering, *J. Biomed Mater. Res. Part A* 104 (2015) 264–271.
- K. Ren, Y. Wang, T. Sun, W. Yue, H. Zhang, Electrospun PCL/gelatin composite nanofiber structures for effective guided bone regeneration membranes, *Mater. Sci. Eng. C* 78 (2017) 324–332.
- V.Y. Chakrapani, T.S.S. Kumar, D.K. Raj, T.V. Kumary, Electrospun cytocompatible polycaprolactone blend composite with enhanced wettability for bone tissue engineering, *J. Nanosci. Nanotechnol.* 17 (2017) 2320–2328.
- O.P. Gautschi, S.P. Frey, R. Zellweger, Bone morphogenetic proteins in clinical applications, *ANZ J. Surg.* 77 (2007) 626–631.
- R. Eivazzadeh-Keihan, A. Maleki, M. de la Guardia, M.S. Bani, K.K. Chenab, P. Pashazadeh-Panahi, et al., Carbon based nanomaterials for tissue engineering of bone: building new bone on small black scaffolds: a review, *J. Adv. Res.* 18 (2019) 185–201.
- G. Mishra, S. Srivastava, B.P. Nagori, Pharmacological and therapeutic activity of *Cissus quadrangularis*: an overview, *Int. J. Pharmtech Res.* 2 (2010) 1298–1310.
- D.K. Deka, L.C. Lahon, A. Saikia, Mukit, Effect of *Cissus quadrangularis* in accelerating healing process of experimentally fractured radius-ulna of dog: a preliminary study, *Indian J. Pharmacol.* 26 (1994) 44–45.
- B.K. Potu, M.S. Rao, N.G. Kutty, K.M.R. Bhat, M.R. Chamallamudi, S.R. Nayak, Petroleum ether extract of *Cissus quadrangularis* (LINN) stimulates the growth of fetal bone during intra uterine developmental period: a morphometric analysis, *Clinics (Sao Paulo)* 63 (2008) 815–820.
- S. Suganya, J. Venugopal, S. Ramakrishna, B.S. Lakshmi, V.R. Giri Dev, Herbally derived polymeric nanofibrous scaffolds for bone tissue regeneration, *J. Appl. Polym. Sci.* 131 (2014) 1–11.
- M. Shin, H. Yoshimoto, J. Vacanti, In vivo bone tissue engineering using mesenchymal stem cells on a novel electrospun nanofibrous scaffold, *Tissue Eng.* 10 (2004) 33–41.
- I. Kalaszczynska, K. Ferdyn, Wharton's Jelly derived mesenchymal stem cells: future of regenerative medicine? Recent findings and clinical significance, *Biomed Res. Int.* 2015 (2015).
- Z.-Y. Zhang, S.-H. Teoh, M.S.K. Chong, J.T. Schantz, N.M. Fisk, M.A. Choolani, et al., Superior Osteogenic Capacity for Bone Tissue Engineering of Fetal Compared with Perinatal and Adult Mesenchymal Stem Cells, *Stem Cells* 27 (2009) 126–137.
- S. Kashte, R.K. Sharma, S. Kadam, Layer-by-layer decorated herbal cell compatible scaffolds for bone tissue engineering: a synergistic effect of graphene oxide and *Cissus quadrangularis*, *J. Bioact. Compat. Polym.* 35 (2020) 57–73.
- S. Kashte, G. Arbade, R.K. Sharma, S. Kadam, Herbally painted biofunctional scaffolds with improved osteoinductivity for bone tissue engineering, *J. Biomimetics, Biomater. Biomed. Eng.* 41 (2019) 49–68.
- H. Shen, L. Zhang, M. Liu, Z. Zhang, Biomedical applications of graphene, *Theranostics* 2 (2012) 283–294.
- Y. Piao, Y. Liu, X. Xie, Change trends of organ weight background data in Sprague Dawley Rats at different ages, *J. Toxicol. Pathol.* 26 (2013) 29–34.
- M.S. Baliga, G.C. Jagetia, J.N. Ulloor, M.P. Baliga, P. Venkatesh, R. Reddy, et al., The evaluation of the acute toxicity and long term safety of hydroalcoholic extract of *Sapthaparna* (*Alstonia scholaris*) in mice and rats, *Toxicol. Lett.* 151 (2004) 317–326.
- Q. Liu, Z. Lei, Q. Wu, I. Awais, M.A.B. Shabbir, The reproductive toxicity of mequindox in a two-generation study in wistar rats, *Front. Pharmacol.* 9 (2018) 1–14.
- A.K. Jaiswal, R.V. Dhumal, S. Ghosh, P. Chaudhari, H. Nemani, V.P. Soni, et al., Bone healing evaluation of nanofibrous composite scaffolds in rat calvarial defects: a comparative study, *J. Biomed. Nanotechnol.* 9 (2013) 2073–2085.
- A.K. Jaiswal, R.V. Dhumal, S. Ghosh, P. Chaudhari, H. Nemani, V.P. Soni, et al., Bone healing evaluation of nanofibrous composite scaffolds in rat calvarial defects: a comparative study, *J. Biomed. Nanotechnol.* 9 (2013) 2073–2085.
- K. Khanna, A. Jaiswal, R.V. Dhumal, N. Selkar, P. Chaudhari, V.P. Soni, et al., Comparative bone regeneration study of hardystonite and hydroxyapatite as filler in critical-sized defect of rat calvaria, *RSC Adv.* 7 (2017) 37522–37533.
- T. Taniguchi, T. Matsumoto, H. Shindo, Changes of serum levels of osteocalcin, alkaline phosphatase, IGF-I and IGF-binding protein-3 during fracture healing, *Injury* 34 (2003) 477–479.
- S.A. Bowles, N. Kurdy, A.M. Davis, M.W. France, D.R. Marsh, Serum osteocalcin, total and bone-specific alkaline phosphatase following isolated tibial shaft fracture, *Ann. Clin. Biochem.* 33 (1996) 196–200.
- S. Bi, P. Wang, S. Hu, S. Li, J. Pang, Z. Zhou, Construction of physical-crosslink chitosan / PVA double-network hydrogel with surface mineralization for bone repair, *Carbohydr. Polym.* 224 (2019), 115176.
- P. Kolar, K. Schmidt-Bleek, H. Schell, T. Gaber, D. Toben, G. Schmidmaier, et al., The early fracture hematoma and its potential role in fracture healing, *Tissue Eng. Part B Rev.* 16 (2010) 427–434.

- [36] T. Kon, T.-J. Cho, T. Aizawa, M. Yamazaki, N. Nooh, D. Graves, et al., Expression of Osteoprotegerin, receptor activator of NF- κ B ligand (Osteoprotegerin Ligand) and related proinflammatory cytokines during fracture healing, *J. Bone Miner. Res.* 16 (2001) 1004–1014.
- [37] P. Martin, D. D'Souza, J. Martin, R. Grose, L. Cooper, R. Maki, et al., Wound healing in the PU.1 null mouse—tissue repair is not dependent on inflammatory cells, *Curr. Biol.* 13 (2003) 1122–1128.
- [38] D. Toben, I. Schroeder, T. El Khassawna, M. Mehta, J.-E. Hoffmann, J.-T. Frisch, et al., Fracture healing is accelerated in the absence of the adaptive immune system, *J. Bone Miner. Res.* 26 (2011) 113–126.
- [39] X. He, Y. Liu, X. Yuan, L. Lu, Enhanced healing of rat calvarial defects with MSCs loaded on BMP-2 releasing chitosan / alginate / hydroxyapatite scaffolds, *PLoS One* 9 (2014) 1–9.
- [40] A. Oryan, S. Alidadi, A. Bigham-sadegh, A. Moshiri, Healing potentials of polymethylmethacrylate bone cement combined with platelet gel in the critical-sized radial bone defect of rats, *PLoS One* 13 (2018) 1–17.
- [41] A.K. Teotia, D.B. Raina, H. Isaksson, M. Tägil, L. Lidgren, Composite bilayered scaffolds with bio-functionalized ceramics for cranial bone defects: an in vivo evaluation Composite bilayered scaffolds with bio-functionalized ceramics for cranial bone defects: an in vivo evaluation, *Multifunct. Mater* 2 (2019) 1–11.
- [42] N. Singh, V. Singh, R. Singh, A. Pant, U. Pal, L. Malkunje, et al., Osteogenic potential of cissus quadrangularis assessed with osteopontin expression, *J. Maxillofac. Surg.* 4 (2013) 52–56.
- [43] M.S. Rao, P. Bhagath Kumar, V.B. Narayana Swamy, N. Gopalan Kutty, Cissus quadrangularis plant extract enhances the development of cortical bone and trabeculae in the fetal femur, *Pharmacologyonline* 3 (2007) 190–202.